Dr. Ludwig M. Böss

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LudwigBoess
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Education & Academic Positions

Jan 2025–now
Chicago, IL

Postdoctoral Researcher, The University of Chicago

with Prof. Dr. Damiano Caprioli

Feb 2024–Dec 2024
Munich

Postdoctoral Researcher, Ludwig-Maximilians-Universität

with Prof. Dr. Klaus Dolag

Apr 2020–Jan 2024
Munich

Astrophysics, PhD, Ludwig-Maximilians-Universität, *magna cum laude*

Thesis: On Simulations of Cosmic Rays in the Large-Scale Structure of the Universe
Supervisors: Prof. Dr. Harald Lesch & Prof. Dr. Klaus Dolag

Sep 2017–Apr 2020
Munich

Astrophysics, M.Sc., Ludwig-Maximilians-Universität

GPA: 4.0
Thesis: Cosmic Rays in Galaxy Clusters - An on-the-fly Fokker-Planck Solver for OpenGadget3; Supervisor: Prof. Dr. Klaus Dolag

Sep 2014–Aug 2017
Munich

Physik, B.Sc., Ludwig-Maximilians-Universität

GPA: 2.7
Thesis: Radial Orbit Instability - Analysis of geometry in unperturbed and perturbed systems; Supervisor: Prof. Dr. Andreas Burkert

Sep 2011–Aug 2014
Munich

Musicology, B.A., Ludwig-Maximilians-Universität

GPA: 3.0
Thesis: Witold Lutoslawski's Concerto for Orchestra in the context of Socialist Realism; Supervisor: Prof. Dr. Wolfgang Rathert

Invited Seminar Talks

December 2025 **Twin Talks - PPPL Seminar**, Princeton, *USA*

November 2025 **Michigan State University Astronomy Seminar**, Lansing, *USA*

October 2025 **THEA Group Seminar**, New York, *USA*

June 2025 **MPIA Galaxy Coffee Seminar**, Heidelberg, *GER*

June 2025 **MPE High-Energy Clusters & Cosmology Group Seminar**, Garching, *GER*

June 2025 **MPA Cosmology Seminar**, Garching, *GER*

December 2024 **LERMA Group Seminar**, Paris, *FR*

December 2024 **IAS Cosmology Group Seminar**, Paris, *FR*

November 2023 **MPE High-Energy Clusters & Cosmology Group Seminar**, Garching, *GER*

June 2023 **CfA Galaxy Cluster Group Seminar**, Boston, *USA*

April 2023 **CCA Plasma Physics Group Seminar**, New York, *USA*

Conferences and Workshops

September 2025 **Relativistic Plasma Physics: From the Lab to the Cosmos**, Santa Barbara, *USA*

June 2025 **Algorithms For Multiphysics Models In The Post-Moore's Law Era**, Los Alamos, *USA*

January 2025 **Particle Acceleration and Transport: from the Sun to Extragalactic Sources**, Rende, *IT (Invited Talk)*

September 2024 **Clusters & Relics II**, Boston, *USA (Invited Talk)*

June 2024 **ICM Theory and Computation Workshop**, Ann Arbor, *USA (Invited Talk)*

October 2023 **Revealing Cosmic Magnetism in the new Future**, Paris, *FR (Talk)*

July 2023 **International Cosmic Ray Conference**, Nagoya, *JP (Talk)*

January 2023 **Cosmic Magnetism in Voids and Filaments**, Bologna, *IT (Talk)*

August 2022 **International Astronomical Union General Assembly**, Busan, *KOR (Poster)*

March 2022 **Clusters & Relics**, Tautenburg, *GER (Talk)*

July 2021 **MIAPP workshop - High energy phenomena in astrophysics**, Munich, *GER (Talk)*

Awarded Computing and Observation Time

2024 **GCS Project (PI)**, Simulations of Cosmic Rays in Galaxy Clusters at Unprecedented Resolution
29 Million CPUh

2023 **GAUSS Large Scale Project (co-I, PI: Klaus Dolag)**, The Local Universe: Galaxies, Clusters, The LSS and Cosmic Rays
69 Million CPUh

2022 **C2PAP Computing Grant (PI)**, Galaxy Clusters with Spectral Cosmic Rays
6.5 Million CPUh

2020 **GAUSS Large Scale Project (co-I, PI: Klaus Dolag)**, COMPASS
50 Million CPUh

2025 **XRISM GO Cycle 2 (Co-PI, PI: Irina Zhuravleva)**, ICM Turbulence in the Merging Chaos of Abell 2256
2 pointings, 290ks total

Student Supervision

2025-now **PhD Thesis**, Student: Anna Ivleva - *assistant supervisor*

2024-now **PhD Thesis**, Student: Daniel Karner- *assistant supervisor*

2023 **Masters Thesis**, Students: Daniel Karner, Julian Sommer - *assistant supervisor*

Skills

Languages

German	First language
English	Fluent
French	Conversation skills
Italian	Basic communication skills

Programming skills

Julia ●●●●○ C/C++ ●●●●○ Python ●●●●○ Fortran ●●●○○
SQL ●●●●○ HTML/CSS ●●○○○

First-Author Publications

in prep. **Entity– Performance-portable Particle-in-Cell Code for Relativistic Plasma Astrophysics. III: Higher order methods**, **Böss, L.**, Vanthieghem, A., Hakobyan, H., Caprioli, D.

submitted **Simulating the LOcal Web (SLOW) - *VI*: γ-ray Emission from the Local Universe**, **Böss, L.**, Khabibullin, I., Karner, D., Dolag, K., Steinwandel, U., Hernández-Martínez, E., Sorce, J. G., arXiv:2510.15634.

2024 **Simulating the LOcal Web (SLOW) - *III*: Synchrotron Emission from the Local Cosmic Web**, **Böss, L.**, Dolag, K., Steinwandel, U., Hernández-Martínez, E., Khabibullin, I., Seidel, B., Sorce, J. G., A&A, 692, id.A232, 25 pp.

2023 **A formation mechanism for 'Wrong Way' Radio Relics**, **Böss, L.**, Steinwandel, U., Dolag, K, ApJL, 957, L16

2023 **CRESCENDO: An on-the-fly Fokker-Planck Solver for Spectral Cosmic Rays in Cosmological Simulations**, **Böss, L.**, Steinwandel, U., Dolag, K., Lesch, H., MNRAS, 519,1, pp.548-572

Co-Author Publications

in prep. **Entity – Performance-portable Particle-in-Cell Code for Relativistic Plasma Astrophysics. I: Curvilinear Special Relativistic Module**, Hakobyan, H., **Böss, L.**, Cai, Y., Chernoglazov, A., Galishnikova, A., Gorbunov, E., et al.

Contributions: Code development, scaling tests on Aurora, shock setup and accompanying section in paper.

submitted **Shock Acceleration in the Intracluster Medium: Implications of Micromirror Confinement**, Diesing, R., **Böss, L.**, Caprioli, D., arXiv:2510.23700.

Contributions: Underlying simulation and interpretation of the results.

submitted **Cosmological Zoom-In Simulation of Odd Radio Circles as Merger-Driven Shocks in Galaxy Groups**, Ivleva, A., **Böss, L.**, Dolag, Klaus, Koribalski, B. S., Khabibullin, I., et al. , arXiv:2507.11781.

Contributions: Student supervision, simulation setup, help with data analysis.

submitted **Magnetic Fields in the Shapley Supercluster Core with POSSUM: Challenging Model Predictions**, Alonso-López, D., O'Sullivan, S., Bonafede, A., **Böss, L.**, et al., submitted to A&A

Contributions: Simulation data, help with interpretation of the results.

submitted **Two intersecting radio shells: relics of galaxy merger shocks?**, Koribalski, B. S., Dolag, Klaus ; Khabibullin, I., **Böss, L.**, et al. , arXiv:2507.11781.

Contributions: Scientific input.

2024 **The glow of axion quark nugget dark matter: II. Galaxy clusters**, Sommer, J. S., Dolag, K., **Böss, L.**, Khabibullin, I., Liang, X., et al., A&A, 691, id.A38, 22pp.

Contributions: Scientific input.

2024 **Towards cosmological simulations of the magnetized intracluster medium with resolved Coulomb collision scale**, Steinwandel, U., Dolag, K., **Böss, L.**, Marin, T., ApJ, 967, 2, id.125, 19 pp

Contributions: Scientific input, Fig. 3, 4, 10 & 13 in publication.

2023 **Insights on the origin of ORCs from cosmological simulations**, Dolag, K., **Böss, L.**, Koribalski, B., Steinwandel, U., Valentini, M., ApJ, 945, 74

Contributions: Modeling of the CR component, related figures and chapters in the paper. Interactivate figure in the online publication.

2022 **Virgo: Scalable Unsupervised Classification of Cosmological Shock Waves**, Lamparth, M., **Böss, L.**, Steinwandel, U., Dolag, K., arXiv:2208.06859

Contributions: Scientific input, development contribution, all figures in publication.

2022 **On the small scale turbulent dynamo in the intra cluster medium: A comparison to dynamo theory**, Steinwandel, U., **Böss, L.**, Dolag, K., Lesch, H., ApJ, 933, 2, 131

Contributions: Analysis tools and scripts. Related test simulations.

2019 **WVTICs - SPH initial conditions for everyone**, Arth, A., Donnert, J, Steinwandel, U., **Böss, L.**, et al., arXiv:1907.11250

Contributions: Implementation of the artificial bias correction and writing the corresponding section. Rerunning the tests and figures for the paper.